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Perspective

Perfect adaptation in biology

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A distinctive feature of many biological systems is their ability to adapt to persistent stimuli or disturbances that would otherwise drive them away from a desirable steady state. The resulting stasis enables organisms to function reliably while being subjected to very different external environments. This perspective concerns a stringent type of biological adaptation, robust perfect adaptation (RPA), that is resilient to certain network and parameter perturbations. As in engineered control systems, RPA requires that the regulating network satisfy certain structural constraints that cannot be avoided. We elucidate these ideas using biological examples from systems and synthetic biology. We then argue that understanding the structural constraints underlying RPA allows us to look past implementation details and offers a compelling means to unravel regulatory biological complexity.

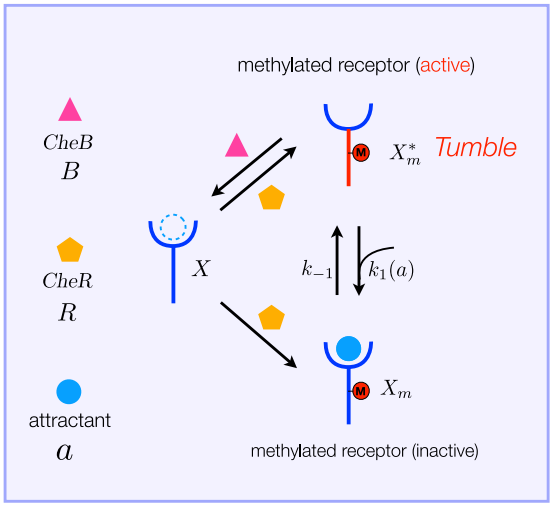
INTRODUCTION

Adaptation is a running theme across all of biology. It is a hallmark of living systems, allowing them to withstand their hostile environments and maintain their function. Claude Bernard's "la fixité du milieu intérieur" (Gross, 1998) and Walter Cannon's homeostasis (Cannon, 1929, 1939) are both manifestations of adaptation strategies employed to maintain a constancy of the internal environment that is resistant to external influences. Central to this adaptation are dynamic regulatory mechanisms that robustly function to counteract influences that are constantly attempting to push the system toward disequilibrium. It is in such regulated environments that living systems can function reliably. Therefore, it is not surprising that much of the complexity of living systems is attributable to the presence of regulatory networks, which are found at all levels of biological organization, from the single cell to the whole organism.

While adaptation is ubiquitous, the requirements for adaptation are not equally stringent for all subsystems. In particular, systems that exhibit homeostasis do not necessarily achieve the type of adaptation requiring variables of interest to be absolutely maintained at a fixed value. Indeed, the word homeostasis, which derives from the Greek language, refers only to maintaining a *similar* stasis, which need not imply maintenance of the *same* stasis. Regulatory strategies that achieve such homeostasis are less restricted and can be found in many flavors across the design space of living systems (Cannon, 1939). In contrast, in certain areas of biological regulation, the requirements for adaptation are much more stringent, necessitating strategies that achieve a perfect adaptation (PA). This implies that a variable of interest must be maintained at a constant value, a setpoint, and must be made to return to that value exactly following persistent disturbances or stimuli that would otherwise change its value to a new level. Systems capable of achieving such an adaptation feat are said to be perfectly adapting. PA phenomena have been widely observed in biological systems at multiple organizational levels and at multiple timescales. These include chemotaxis in *E. coli* (Barkai and Leibler, 1997; Alon et al., 1999) and *Dictyostelium discoideum* (Levchenko and Iglesias, 2002), intracellular osmolarity (Muzzey et al., 2009; You et al., 2012), sigma-70 activity (Briat et al., 2016a), ammonia uptake (Mairet, 2018), temperature control (Ni et al., 2009), copper and zinc homeostasis (Mairet, 2018), iron homeostasis (Ni et al., 2009), calcium homeostasis (El-Samad et al., 2002), and glucose uptake in growing cancer cells (Tveit et al., 2020), among several others.

In this perspective, we will focus almost exclusively on robust perfectly adapting systems. As we will see, such systems have strict constraints on their network architectures. Elucidating these constraints and their role in achieving RPA will shed light on those structural aspects of a network that may change or evolve without loss of adaptation, while pointing to others that cannot be altered without destroying adaptation. Intriguingly, the search for such network features will lead us to a relatively small set of subnetwork motifs whose topological features are responsible for conferring adaptation to a much larger complex network in which they may be embedded. When the dynamics are significantly noisy, as often occurs in cellular networks, new results show that the class of networks achieving robust perfect adaptation (RPA) is further constrained. Understanding the topological constraints of these subnetworks will lead to a deeper understanding of how complex biological systems achieve and maintain robust adaptation.

We start by looking at two case studies that exhibit RPA. This will motivate the mathematical characterization of robustly adapting systems, which yields algebraic conditions for adaptation. Next, we look for biochemical network topologies that inherently satisfy these conditions. Such topologies include both integral feedback and incoherent feedforward topologies. We next examine RPA in noisy stochastic networks and characterize all network topologies that achieve RPA in the presence of noisy dynamics. Finally, we discuss how our understanding of RPA in endogenous networks can inform our design of synthetic biological circuits that possess desirable adaptation features, and we highlight some recent successes in engineering such circuits and their potential applications.



Dynamics:

$$\frac{dX_m}{dt} = -k_{-1}X_m + k_1(a)X_m^* + \alpha V_R R$$

$$\frac{dX_m^*}{dt} = k_{-1}X_m - (k_1(a) + k_B B)X_m^* + (1-\alpha)V_R R$$

$$X_T := X_m + X_m^*$$

$$\frac{dX_T}{dt} = V_R R - k_B B X_m^*$$

Perfect Adaptation:

$$\text{At steady-state, } X_m^* = \frac{V_R R}{k_B B} \textit{ independent of } a$$

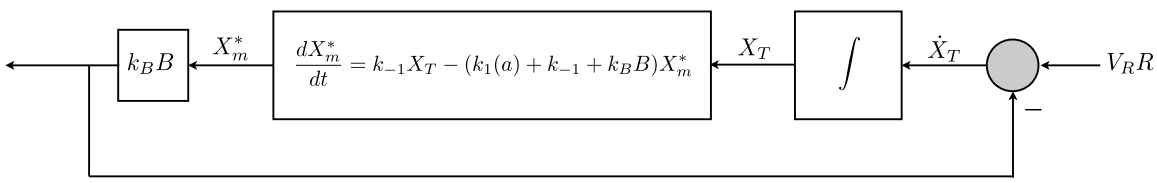


Figure 1. Bacterial chemotaxis

A simplified model of bacterial chemotaxis, adapted from Alon (2019). The model contains the essential elements for achieving RPA. The dynamics are described by two variables, X_m and X_m^*. The dynamics of the sum of the two variables, X_T, has an integral structure that imposes perfect adaptation (PA). The dynamics can be formulated in terms of a closed-loop dynamical system with an integrator element appearing in feedback with the dynamics of X_m^*.

RPA IN BIOLOGY: SOME CASE STUDIES

Here we will explore two of the example systems at very different scales in order to bring out the underlying structural mechanisms that enable RPA in these systems. One of these is chemotaxis in *E. coli*, which is an example of RPA at the intracellular level; the other is plasma calcium homeostasis, which demonstrates PA at the whole-organism level.

Chemotaxis in *E. coli*

We start with bacterial chemotaxis adaptation (Barkai and Leibler, 1997; Alon et al., 1999; Yi et al., 2000). Bacteria respond to stimulants such as chemo-attractants (or chemo-repellents) by moving toward (or away) from these stimulants. In *E. coli*, this is achieved through a biased random walk, consisting of alternating episodes of straight line runs and random angle-changing tumbles (Berg and Brown, 1972). When an attractant is first added uniformly, the frequency of tumbles is sharply reduced, but then it adapts to its level prior to adding the attractant. This sensory adaptation is exact (or perfect), and it enables the cell to respond to a large range of concentrations, in effect increasing the dynamic range of concentrations to which it can respond. We will describe a simplified model of bacterial chemotaxis adapted from Alon (2019).

The methylated receptor and the kinase to which it is bound can either be in an active state, X_m^*, or an inactive state, X_m (Figure 1). In its active state, it phosphorylates a response regulator, which in turn binds to the flagellar motor, changing its rotation direction and promoting tumbling. Thus, the concentration of X_m^* (the activity) dictates the tumbling rate, which is the quantity that adapts to the concentration of the chemo-attractant, and therefore the output of our system. The receptor is most active when it is not bound to a chemo-attractant molecule (ligand). Once the receptor is bound to a ligand, the probability of deactivation increases, so the rate of conversion from X_m^* to X_m is an increasing function of the ligand concentration, a. The mechanism thus described does not lead to adaptation of X_m^* but would result in less activity (and hence lower tumbling frequency) the higher the ligand concentration. Adaptation is achieved when another mechanism that demethylates the receptor through a phosphatase, CheB, is introduced. This demethylation process deactivates the receptor. We will refer to the unmethylated receptor by X, regardless of its binding state to the ligand. The two key assumptions required for RPA are that (1) CheB only demethylates the active receptor and (2) methylation of the unmethylated receptor X by the enzyme CheR is a zero-order process, i.e., the enzyme functions in its saturation regime. Following Alon (2019), we adopt a simple second-order model of the chemotaxis process (see Figure 1). It can be seen from the model that the dynamics of X_T, defined as the sum of the concentrations of X_m and X_m^*, is given by $\dot{X}_T = V_R R - k_B B X_m^*$; hence, at steady state, the activity is given by $X_m^* = V_R R / k_B B$. This last equation shows clearly that the activity does not depend on the concentration of the chemo-attractant, a, nor does it depend on the specific values of k_1 or k_{-1}. RPA is thus achieved for the output $X_m^* = \frac{V_R R}{k_B B}$.

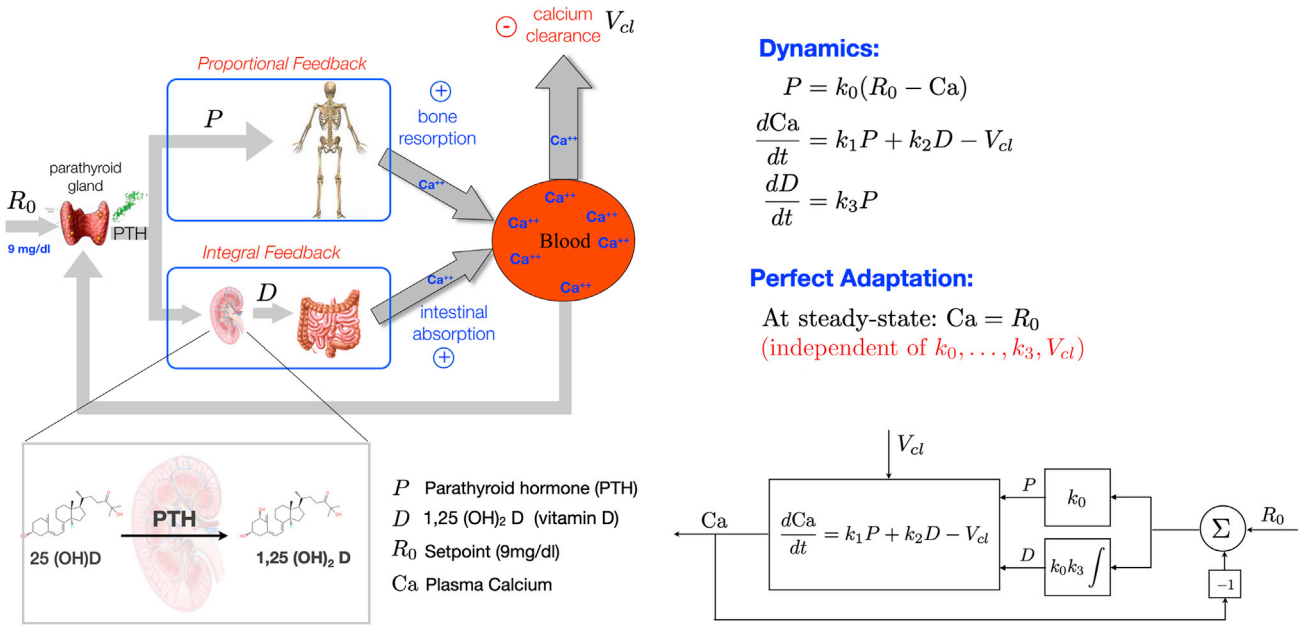


Figure 2. Calcium homeostasis

The essential regulatory pathways responsible for mammalian calcium homeostasis. Two sources of calcium, intestinal absorption and bone resorption, work in concert to compensate for loss of blood calcium. The regulation of intestinal absorption introduces an integral feedback element that is responsible for the system achieving RPA. A block diagram shows exactly how the integral feedback acts on the closed-loop dynamics.

Calcium homeostasis

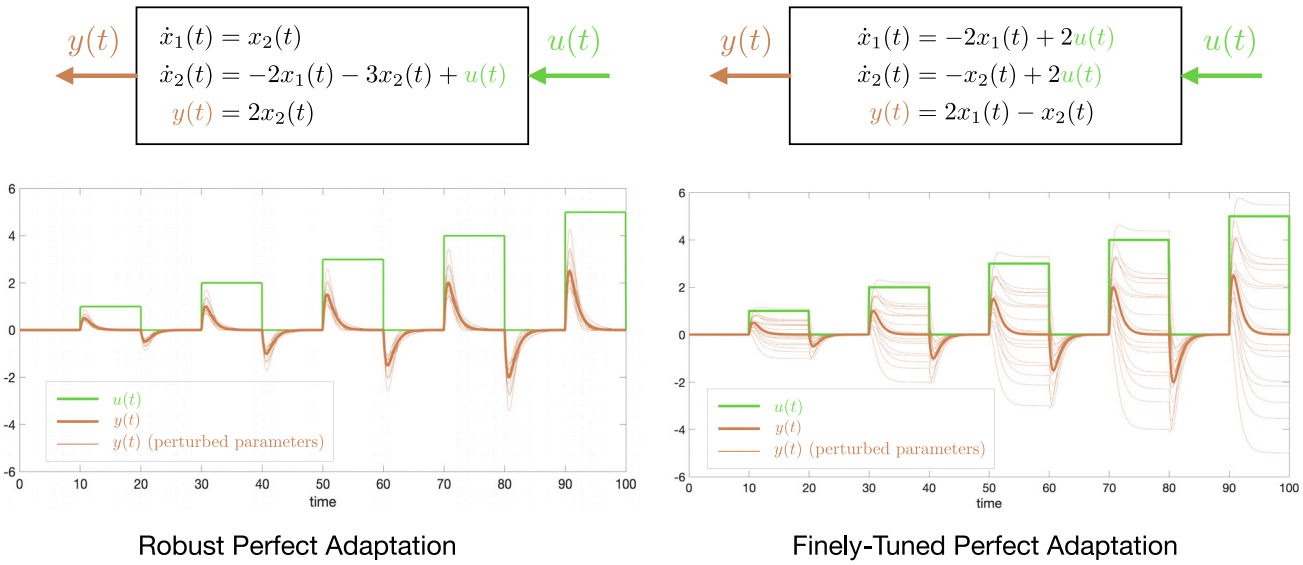
For our second case study, we will examine calcium homeostasis in mammals, which functions at a very different scale from chemotaxis. In mammals, calcium ions, Ca^{++}, play multiple essential biochemical roles, as they are central to muscle function, nerve function, blood clotting, and cellular signaling, among other functions. For this critical biochemical role of calcium to be achieved, calcium concentration must be tightly regulated in the blood plasma of mammals. Nature has evolved exquisite regulatory mechanisms that maintain calcium concentrations within tight limits through RPA. It is well known that blood plasma calcium concentrations are maintained close to 9 mg/dL, in spite of large disturbances to the calcium flux. In particular, during lactation, a high outflux of calcium from the blood is observed, due largely to the removal of calcium by the mammary glands. This large calcium clearance persists throughout the period of lactation, which lasts many months. In spite of that, calcium concentrations in the blood are easily maintained at physiological levels. There are two influx sources of calcium that compensate for the blood calcium clearance due to milk production: bone resorption and intestinal absorption. These calcium fluxes are coordinated to regulate calcium concentrations in the blood robustly and perfectly. We briefly describe the feedback mechanisms behind this regulation. Blood concentration of calcium is monitored by the parathyroid gland, which secretes into the blood parathyroid hormone (PTH) in direct proportion to the calcium deficiency. Therefore, blood concentration of PTH can be viewed as a good reporter for the deviation of blood calcium from its set point of 9 mg/dL. Bone resorption of calcium is stimulated by PTH, so this calcium flux is proportional to the calcium deviation from its set point.

Intestinal absorption of calcium, on the other hand, is highly stimulated by $1,25(\mathrm{OH})_2\mathrm{D}$, an active form of vitamin D. Activation of this hormone is achieved by hydroxylation of an inactive form of vitamin D, namely $25(\mathrm{OH})\mathrm{D}$, a process that takes place in the kidney. Critical to perfect adaption is that the hydroxylation by PTH is a zero-order process, which is guaranteed due to the large abundance of the inactive substrate. Together, bone resorption and intestinal absorption work in concert to achieve calcium homeostasis and, in fact, robust perfect adaption. A simple model of this process is shown in Figure 2. From the model it can be seen that $\dot{D} = k_0 k_3 (R_0 - \mathrm{Ca})$, which implies that at steady state, the blood calcium concentration Ca is equal to its setpoint, R_0. This adaptation is independent of the parameters k_0, k_1, k_2, k_3 or the disturbance V_{cl}. In fact, it is also robust to dynamic perturbations. Like bacterial chemotaxis, RPA here is achieved for $\mathrm{Ca} = R_0$.

In a later section, we examine the common structures that enabled RPA in both bacterial chemotaxis and calcium homeostasis, despite the vastly different scales at which they function.

PA: ROBUST VERSUS FINELY TUNED

Motivated by the above examples, we would like to tackle PA in more general settings. We start with a given dynamical system that represents a biological system of interest. Adaptation refers to a particular variable, an output of interest (which we refer to as y), and its resistance to changes in another system input variable, which acts as an input or disturbance (referred to as u). Therefore, when we say the system achieves adaptation, it implies that we have in mind a particular system input and a particular system output that is resistant to changes in the input. Here



Robust Perfect Adaptation

Finely-Tuned Perfect Adaptation

Figure 3. Robust versus finely tuned adaptation

The two systems in the figure achieve PA for the parameters indicated, as can be seen from the solid brown trajectory. In fact, both systems are dynamically equivalent from an input-output point of view. However, the system on the right loses PA whenever any of its nonzero parameters are perturbed, no matter how slightly. For this system, PA is finely tuned. In contrast, the system on the left maintains PA for any perturbation of the nonzero parameters, and hence achieves RPA. Thin brown trajectories show output trajectories of both systems under a random change of each of the parameters (within 50% of their shown values).

we will be interested in PA, so that when the input changes from any constant level to any other, the output will return to its pre-change value after some transient excursions from that value. If, in addition, PA is maintained in spite of changes in specified system parameters/dynamics, we say the system achieves RPA; otherwise the PA is said to be "finely tuned."

Consider the two systems in Figure 3, both of which achieve PA. Indeed, for any constant input u, the output y at steady state will be equal to zero. This can also be seen from the the output trajectory simulations in response to an input consisting of an increasing train of pulses. Not only do the two systems display the same constant output at steady state, their transient dynamics are also identical. In fact, from an input-output viewpoint, these two systems are indistinguishable from one another. However, there is an important difference between them. While perturbing any (or all) of the nonzero parameters (ordinary differential equation [ODE] coefficients) of the system on the left leaves the PA property intact, perturbing any of the nonzero parameters of the system on the right will lead to the loss of the PA property. This can be seen from the trajectory simulations in response to the pulse-train input, when applied to several systems obtained by making small random perturbations to the nonzero system parameters. The system on the left therefore achieves RPA, while the PA of the system on the right is finely tuned.

Mathematical characterization

We now obtain conditions for a system to achieve PA. Consider a general description of a biochemical network:

$$\dot{x}(t) = f(x,u)$$

$$y(t) = g(x)$$

where x is an n dimensional vector of the network variables and u and y are the input and output of interest, respectively, relative to which PA is considered. For simplicity, both u and y are taken to be scalar. We assume that for every constant input u there exists a fixed point, x_u, for the dynamics and that this fixed point is dynamically stable. Based on our definition above, PA is achieved for this network if for every constant u there exists a constant k that does not depend on u, such that the following steady-state equations are satisfied:

$$0 = f(x_u, u) \qquad \text{(Equation 1)}$$

$$k = g(x_u). \qquad \text{(Equation 2)}$$

Clearly, k is the value to which the output adapts at steady state. Equations 1 and 2 are satisfied if and only if for every constant u, the linear zero-sensitivity equations are solvable, i.e.:

$$\begin{matrix} 0 & = & A_u x + b_u \\ 0 & = & c_u x \end{matrix} \quad \text{if and only if} \quad \det\begin{bmatrix} A_u & b_u \\ c_u & 0 \end{bmatrix} = 0$$

where $A_u = \frac{\partial f}{\partial x}(x_u, u)$ is the $n \times n$ Jacobian matrix, $b_u = \frac{\partial f}{\partial u}(x_u, u)$ is an $n \times 1$ vector, and $c_u = \frac{\partial g}{\partial x}(x_u)$ is a $1 \times n$ vector. Note that because of the stability condition on the fixed point for every u, the eigenvalues of A_u will have negative real parts, and hence A_u will be nonsingular. Therefore, if the zero-sensitivity equations have a solution, x, it will be unique. When inspecting the derivative of the steady state Equations 1 and 2, one can further conclude that such a solution, x, must be equal to $\frac{\partial x_u}{\partial u}$.

For a linear dynamical system, $f(x,u) = Ax + bu$ and $g(x) = cx$, and therefore the matrices in the determinant will not depend on u and the determinant test involves a single matrix. The

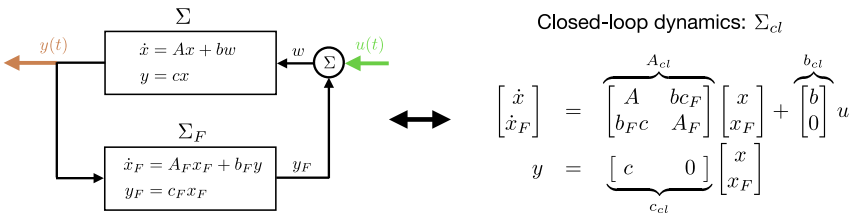


Figure 4. The feedback interconnection of two linear systems, Σ and Σ_F

y, the output of Σ, is fed as an input to the system Σ_F. w, the input of Σ, is the sum of y_F and the external input, u. The resulting closed-loop system, Σ_{cl}, with input u and output y, is a linear system with dynamics derived from the dynamics of Σ and Σ_F, as shown on the right.

determinant condition above gives us a simple algebraic condition for PA (see also Yi et al., 2000). This condition is equivalent to the so-called co-factor conditions (Tang and McMillen, 2016) and to the "RPA equation" given in Araujo and Liotta (2018).

As a quick application, this determinant condition above is satisfied for both systems, as seen in Figure 3. As a function of the nonzero parameters of the system on the left, the determinant remains zero after perturbing these parameters, indicating the robustness of PA. On the other hand, the determinant is very sensitive to perturbations of the nonzero parameters of the system on the right. Indeed, changes to any of these parameters, no matter how small, make the determinant nonzero, thereby signaling the loss of PA for the corresponding perturbed system and underscoring the finely tuned nature of PA in the unperturbed system.

Connections with integral control

The PA property is related to the existence of a function of the state variables that can be expressed as an integral of the output only. More precisely, we say that a dynamical system with output of interest y and state variables $x(t)$ employs integral control, if for a scalar function $h(\cdot)$, $\dot{h}(x)$ depends only on the output y, i.e., we have $\dot{h}(x) = f(y)$ for some function f. We also require that $f(y) = 0$ has a unique solution. With this definition, the connection of integral control with PA should become clear: if the dynamical system is stable, then at steady state its variables x converge to a fixed point, x_u, which in turn implies $\dot{x} = 0$, from which it follows that $f(y) = \dot{h}(x) = 0$. This means that at steady state y takes on a unique constant value independent of any constant input u. If this property is maintained for admissible perturbations of the dynamical systems itself, then PA will be robust. Having $\dot{h}(x) = f(y)$ can be interpreted as having a function of the output $f(y)$ driving the state $z = h(x)$ through a pure integrator.

For a linear system, the existence of integral control is necessary and sufficient for PA, and in this case a linear function h can always be found. Indeed, if the determinant condition above is satisfied, there exists a nonzero vector α such that $\alpha^T[A \quad b] = [c \quad 0]$. Taking $h(x) = \alpha^T x$, we can see that $\dot{h}(x) = \alpha^T(Ax + bu) = cx = y$. As an example, for the system in Figure 3 (left), if we define $z = 2x_1$, then $\dot{z} = y$ and $z(t) = \int_0^t y(s)ds$, which will be true regardless of the input y. In this form, it is easy to see that $y(t)$ must go to zero for any u; otherwise $z(t)$ will not tend to a constant in steady state, as required by the stability of the fixed point. We finally point out that for a nonlinear system that achieves PA, a linear integral control function, h_u, will exist for each linearization.

INTEGRAL FEEDBACK: AN EFFECTIVE STRATEGY FOR RPA

It is instructive to explore the role of integrators when they appear in feedback interconnections of networks. Focusing on linear systems, consider two subsystems, Σ and Σ_F, in feedback with one another, as shown in Figure 4. Having an integrator, Σ_F, in the feedback subsystem means that for some scalar function, h, $\dot{h}(x_F) = f(y)$. Assuming closed-loop stability, this implies that at steady state $f(y) = 0$ regardless of u, which in turn implies PA. Since this condition does not depend on the dynamics of the system in the forward path, Σ, this also implies PA will be achieved for any perturbation in Σ that does not destabilize the system. Thus, integral feedback implies RPA. In fact, any perturbation in Σ_F that does not disturb the integrator structure and maintains closed-loop stability will also result in PA as well. This argument can be made even when the two subsystems are nonlinear. So indeed, having an integrator structure in the feedback subsystem is sufficient for RPA for general feedback interconnections. The chemotaxis and calcium homeostasis networks considered above both admit an integral feedback structure. Figures 1 and 2 show the dynamics of these systems organized to isolate the integral feedback structure for each, highlighting the way it impacts the rest of the system.

The above analysis gives us a powerful strategy for verifying or even designing RPA in general biochemical network settings. If one can identify simple biochemical networks that realize integration, then such networks will deliver RPA whenever they are interconnected in feedback with arbitrary biological networks. Such adaptation will be robust to uncertainty and to perturbations in the arbitrary network as well as to constant disturbances, so long as the closed-loop interconnection remains dynamically stable. This has obvious, important implications for both systems biology and synthetic biology. Over the past two decades, several such integral structures have been identified, and others continue to be discovered.

Figure 5 shows some very simple examples of biomolecular motifs that realize integration. These involve one-species motifs such as the autocatalytic integral motif (Drengstig et al., 2012; Shoval et al., 2011; Briat et al., 2016b), the inflow- and outflow-driven zero-order integral motifs (Ni et al., 2009), and the P-type and N-type antithetic integral motifs (Briat et al., 2016a). Note that in all these examples RPA will be achieved. In particular, the output $y = X_L$ goes to μ/θ at steady state for arbitrary networks and constant inputs. While these are specific examples of integral feedback motifs, they are not the only ones. In Xiao and Doyle (2018), a systematic approach is introduced whereby integral variables that are implementable with chemical reaction networks are found, thereby generating new motifs that realize RPA.

To further build intuition regarding the role of integral feedback in achieving RPA, we take a closer look at one of the motifs in Figure 5. The N-type antithetic feedback network comprises two species, Z_1 and Z_2. Here, Z_1 is expressed constitutively, while the expression of Z_2 depends on the network variable

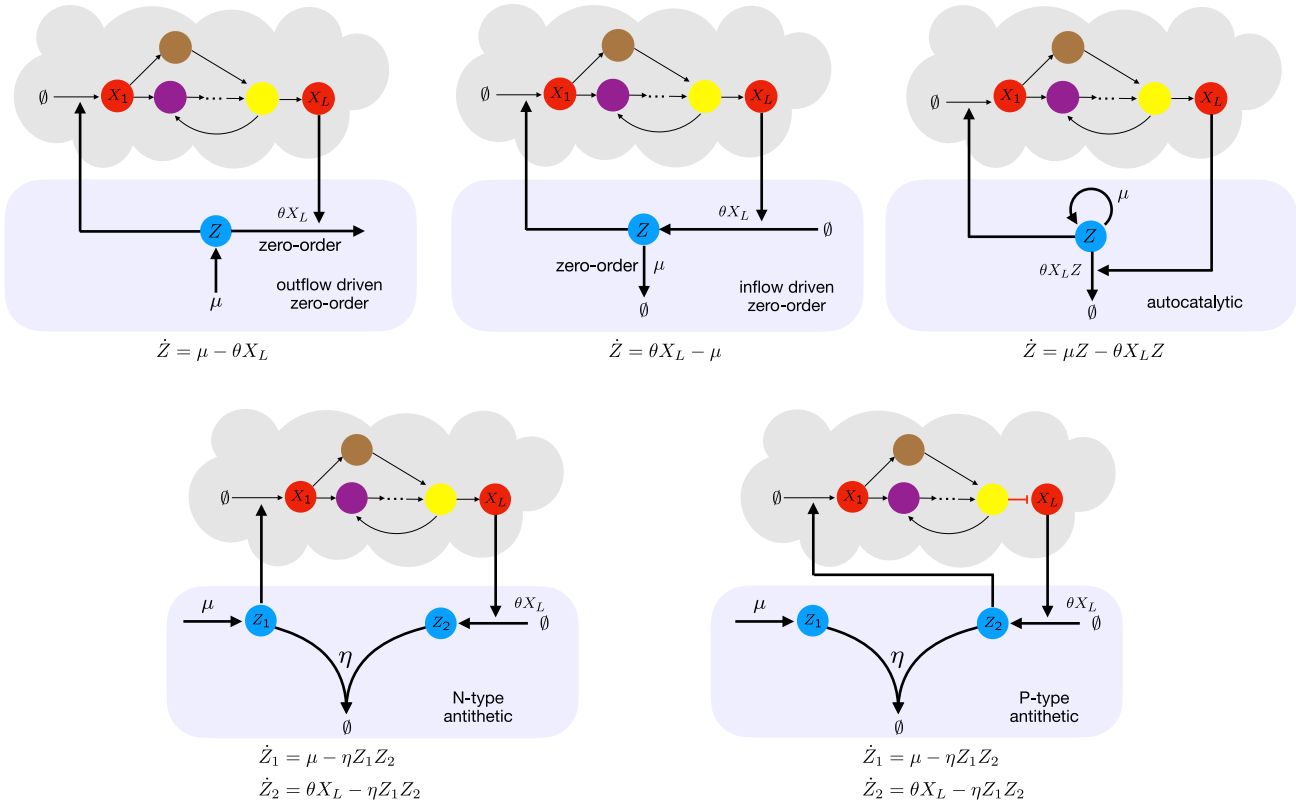


Figure 5. Biomolecular integral feedback motifs

Several biochemical reaction motifs giving rise to integral feedback. Once interconnected with a complex network (clowd), and provided the closed-loop system is stable, the output X_L will go to μ/α in the steady state, achieving RPA for any constant input introduced in the complex network and for any perturbation that does not change the stability of the interconnection.

that needs to achieve RPA, namely Z_L; the reaction producing Z_2 can be thought of as a sensor of the concentration of X_L. Z_2 in turn induces the production of X_1, which drives a network that is assumed to lead to the activation of X_L. The scenario when X_1 leads to the *repression* of X_L can be handled similarly by the $N-type$ antithetic controller. A third reaction combines one molecule of Z_1 with a molecule of Z_2 and leads to their mutual annihilation (or simply their sequestration into an inactive complex). This reaction implements the antithetic motif, which brings about negative feedback regulation that also realizes mathematical integration. To get an insight into the regulatory action of the antithetic motif, imagine that X_L deviates (positively) from its desired steady-state value due to some perturbation. This will result in an increase in the rate of production of Z_2, which increases the rate of the antithetic reaction, leading to a decrease in the concentration of Z_1, followed by a subsequent decrease in the concentration of X_L, which counteracts its increase and brings it back into its steady-state value before the perturbation. That this adaptation is perfect and robust is the result of the integral action of the feedback. To see this, consider the dynamics of Z_1 and Z_2:

$$\dot{Z}_1 = \mu - \eta Z_1 Z_2$$

$$\dot{Z}_2 = \theta X_L - \eta Z_1 Z_2.$$

Subtracting these equations gives us the dynamics of $Z_1 - Z_2$ and reveals the integrator

$$\dot{Z}_1 - \dot{Z}_2 = \mu - \theta X_L \quad \text{or equivalently}$$

$$(Z_1 - Z_2)(t) = \int_0^t (\mu - \theta X_L(s))ds.$$

Dynamic stability dictates that $(Z_1 - Z_2)(t)$ approaches a constant value at steady state. The only way this can be achieved is when the integrand approaches zero, i.e., $X_L(t)$ converges to μ/θ, which is precisely the value to which X_L adapts, regardless of the dynamics of the network. Thus, the integral action brings about an adaptation that is both perfect and robust to network perturbations.

Integral feedback is necessary for RPA for linear feedback networks

We revisit our feedback system in Figure 4, where a linear process, Σ, is in feedback with another linear process, Σ_F, resulting in the closed-loop system, Σ_{cl}, having input u and output y. Suppose RPA is achieved for this interconnection. Here we will be explicit about the class of perturbation for which PA is robust. In particular, RPA means PA is achieved for all systems, Σ_δ, in a neighborhood of Σ, i.e., for those with $\dot{x} = A_\delta x + B_\delta w$; $y = c_\delta x$, where $\|A - A_\delta\| < \varepsilon$, $\|b - b_\delta\| < \varepsilon$, and $\|c - c_\delta\| < \varepsilon$, for some

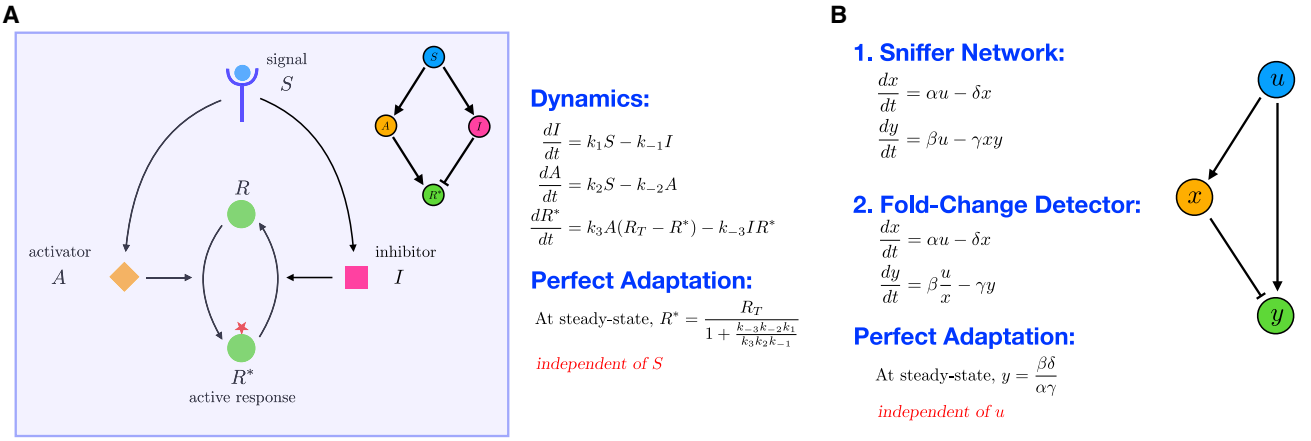


Figure 6. IFFLs can achieve RPA

(A) A simple signaling model for eukaryotic gradient sensing that has been studied in the context of chemotaxis in *Dictyostelium discoideum* and neutrophils (Levchenko and Iglesias, 2002). The model realizes an IFFL and exhibits RPA with respect to S. While the value of the output R^* depends on the network parameters, adaptation to S for a given choice of these parameters does not.

(B) Two systems that use IFFLs to achieve RPA, the sniffer system (Tyson et al., 2003) and a fold-change-detection system (Goentoro et al., 2009; Shoval et al., 2010, 2011).

ε>0. We now show the remarkable fact that integral feedback is in fact *necessary* for RPA. By this we mean that if RPA is achieved, there will be some linear combinations of the state derivatives $z = \alpha \dot{x}_F$ that depends only on the output y.

Suppose the interconnected system achieves RPA. Then it achieves PA for some Σ_δ, satisfying $c_\delta A_\delta^{-1} b_\delta \neq 0$. Let $\Sigma_{cl} = (A_{cl}, b_{cl}, c_{cl})$ be the resulting closed-loop system from the interconnection of Σ_δ with Σ_F. This means that for any constant input u, $y_{cl} = 0$ at steady state. Therefore $0 = A_{cl}x + b_{cl}$ and $0 = c_{cl}x$ for some $x = [x_1 \quad x_2]^T \neq 0$. Substituting for A_{cl}, b_{cl}, and c_{cl}, this means

$$0 = A_\delta x_1 + b_\delta c_F x_2 + b_\delta, \quad 0 = b_F c_\delta x_1 + A_F x_2 \quad 0 = c_\delta x_1.$$

Multiplying the first equation by $c_\delta A_\delta^{-1}$ and using the third equation gives $x_2 \neq 0$. At the same time, the second and third equation imply that $A_F x_2 = 0$, which in turn implies that A_F is singular, and hence for some $\alpha \neq 0$, $\alpha^T A_F = 0$. Thus for the linear combination of the states of the feedback system given by $z = \alpha^T x_F$, $\dot{z}$ is solely a function of the feedback system's output y, which implies the presence of an integrator in the feedback system Σ_F.

These results are a special case of what is referred to in control theory as the internal model principle (Francis and Wonham, 1976). This principle stipulates that a feedback system that robustly rejects an input signal of a particular type (e.g., constant input, a sinusoidal input, etc.) must be able to produce a copy of that signal, i.e., the internal dynamics need to encode an model of that signal in the feedback path. In the context of our example, for RPA to be achieved, y_F must be able to produce any constant in the steady state, in spite of y being zero. This implies that the feedback system Σ_F must have an internal model capable of producing constants. Clearly, such a model must be an integrator. While the above necessity result is shown here for linear systems, generalizations for nonlinear systems under suitable conditions has been reported (see Sontag, 2003; Huang et al., 2018b and the references therein).

RPA WITH INCOHERENT FEEDFORWARD LOOPS

An incoherent feedforward loop (IFFL) is a regulatory motif that connects an input to an output of interest through two paths with opposing actions, with one path activating or enhancing the output and the other inhibiting it. Although seemingly contradictory, this combined effect on the output is best understood when one looks at the transient dynamics, as one path is often delayed through the presence of intermediate nodes. IFFLs are prevalent in biology (Alon, 2019), and they play a particularly important role in signaling pathways (Ferrell, 2016). For example, they are involved in EGF to ERK activation (Sasagawa et al., 2005), glucose to insulin release mechanisms (Menè et al., 1997), ATP to intracellular calcium release (Mahaut-Smith et al., 2000), nitric oxide to NF-κB activation (Ridnour et al., 2007), and microRNA regulation (Tsang et al., 2007), among other functions. In steady state, IFFLs are capable of achieving PA. One example of such an adaptation was proposed (Levchenko and Iglesias, 2002) as a model for eukaryotic gradient sensing with applications to chemotaxis in *Dictyostelium discoideum* and neutrophils (Figure 6A). The figure shows a very simple signaling pathway in which S is a signal that is transmitted downstream to activate an activator, R, through two paths. The model is based on the assumption that S increases the concentration of an activator, A, which converts a response element, R, into its active form, R^*. At the same time, an inactivator, I, mediates the reverse conversion of R^* to R. As can be seen from the simple model (Figure 6A), this mechanism leads to PA, whereby the concentration of R^* does not depend on the signal S in the steady state. Critical to this formulation is that the activation of A and of I exhibit similar functional dependence on S, e.g., both are linear functions of S. Note that the PA property exhibited here is robust to changes in the network parameters, even though changes in such parameters affect the steady state value of the output. In this sense, we can say these IFFL systems achieve RPA.

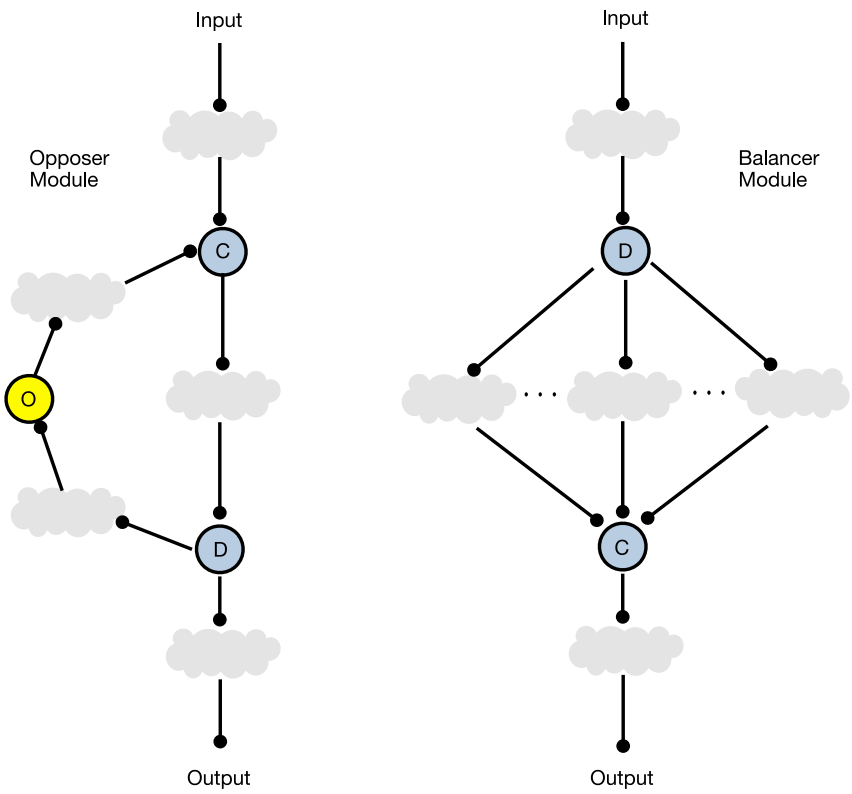


Figure 7. Two module classes form the topological basis of RPA networks

Any RPA-achieving network consists of a suitable interconnection of basis modules in the classes shown (Araujo and Liotta, 2018). The basis module on the left is referred to as an "opposer module" and consists of one or more opposer nodes in the feedback path. The second basis module, shown on the right, is referred to as a "balancer module" and consists of multiple parallel paths connecting the input to the output through a connector node, *C*. The clouds refer to arbitrary networks. The opposer module performs integral action at the opposer nodes, which always appear in the feedback path, whereas the balancer module combines multiple paths from the input to the output such that their impact balances perfectly.

Additional IFFL circuit examples that lead to RPA include the "sniffer" IFFL circuit (Tyson et al., 2003) and an IFFL circuit for fold-change detection (Goentoro et al., 2009; Shoval et al., 2010, 2011), both of which are shown in Figure 6A. The two circuits differ in the inhibition mechanisms, with one enhancing degradation of the output molecule y and the other repressing its creation (Shoval et al., 2011). In both cases, the value of y at steady state depends only on the network parameters, not on the input u.

THE TOPOLOGY OF RPA-ACHIEVING NETWORKS

We have seen two basic mechanisms for generating RPA, integral feedback and the incoherent feedforward, and showed their function on fairly simple circuits. We have also seen that, given a complex network with an output of interest that needs to be robustly and perfectly regulated, such robust and perfect regulation can be achieved by interconnecting the network with an integrator in a feedback fashion. Provided closed-loop stability is maintained, the variable of interest will have the RPA property. We will later see that this has tremendous use for synthetic implementation. The question we address here is a little different. What are the topological features of networks that achieve RPA? In other words, given a complex network with one input and one output of interest that achieves RPA with respect to that input, what is the structure of all network topologies that can achieve such RPA? Can these be characterized?

This question was explored in Ma et al. (2009) for 3-node networks using high-throughput computational searches. In this setting, the authors found that only two types of signaling motifs produced networks implementing RPA: a negative feedback loop with a buffering node and an IFFL with a proportioner node.

A computational approach of this type does not lend itself to larger networks, due to the explosion of computational requirements, and a different approach is needed for such networks. The problem of identifying the topological requirements of biological networks to achieve RPA was recently addressed in Araujo and Liotta (2018). Starting with a simple algebraic condition for RPA based on the determinant of a suitable minor of the system's Jacobian matrix, the authors painstakingly expand this determinant condition to uncover the topological structure that any RPA-achieving network must possess. They establish that two well-defined classes of network topologies span the space of all possible RPA-capable networks, regardless of size. Any RPA-achieving network consists of a suitable interconnection of basis modules in these classes. The topology of these basis modules is shown in Figure 7. One basis module is referred to as an "opposer module" and consists of one or more opposer nodes in the feedback path. The second basis module is called a "balancer module" and consists of multiple parallel paths connecting the input to the output through a connector node, *C*. The opposer module performs integral action at the opposer nodes, which always appear in the feedback path (see Figure 7). The balancer module allows multiple paths from the input to the output so that their impact balances perfectly and the output becomes independent. It should be clear that these modules are elaborations of the integral feedback and incoherent feedforward strategies discussed earlier. The interesting aspect of this analysis is that all RPA networks, regardless of size and complexity, consist of suitable interconnections of these basic modules. We finally note that although the dynamics of an incoherent feedforward topology can be formulated in a way to show that PA is achieved through integral action, topologically it is quite distinct from the integral feedback topology, and therefore both topologies are required to span the entire set of RPA networks.

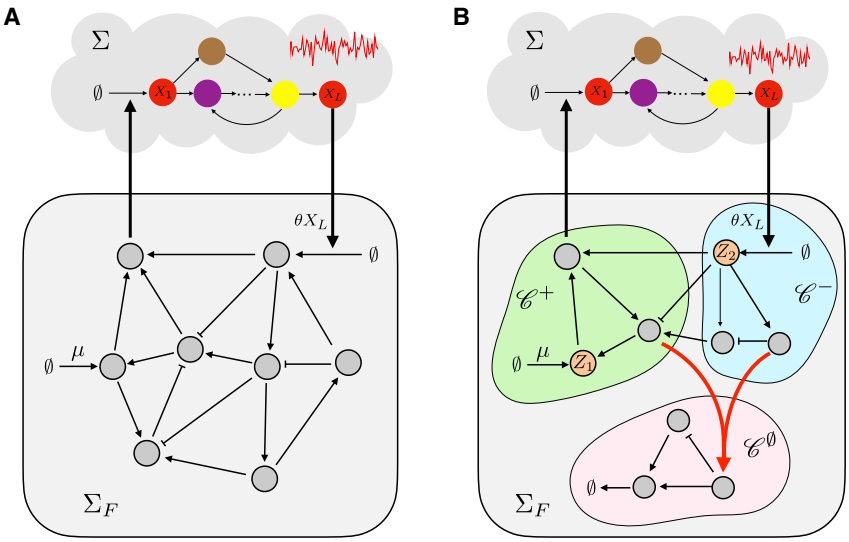


Figure 8. RPA in the presence noisy dynamics
(A) An arbitrary complex network, Σ, interconnected with a biochemical controller network with the goal of having the mean value of the output of interest, $\mathbb{E}(X_L)$, converge to μ/θ in the steady state despite constant disturbances and dynamic perturbations in the process, Σ, and noisy dynamics affecting the entire feedback connection. These properties define RPA in the stochastic setting.
(B) The topology of all feedback interconnections that achieve RPA. It can be seen that every controller that achieves RPA in the stochastic setting must embed the antithetic motif (Aoki et al., 2019).

RPA FOR NOISY DYNAMICS

The internal cellular environment is abuzz with noise, arising from the random nature of biochemical reactions at the molecular level. This randomness affects the timing and order of biochemical reactions and manifests as fluctuations in the copy number of cellular species. The impact of these fluctuations is particularly noticeable when species exist in low copy numbers. These fluctuations propagate through the dynamics of cellular reactions and can have an important impact on the expected values of downstream species as well as their variability. To understand how these stochastic effects impact cellular function, one must study the stochastic dynamics of intercellular networks (Anderson and Kurtz, 2015). Of interest to us here is to understand how these dynamics impact RPA and whether certain architectures are more suitable than others when trying to achieve RPA for systems with stochastic dynamics.

In the presence of large stochastic fluctuations, some of the biomolecular controller motifs in Figure 5 may not function as well as in the deterministic (or noise-free) setting. For example, in the presence of such fluctuations, the assumptions made to realize zero-order degradation reactions may no longer apply, or they may apply only intermittently, making the degradation rate dependent on Z and breaking PA. For the autocatalytic motif, the population of Z will go to zero with probability one in the steady state, regardless of its initial population. In other words, the system will have an absorbing state, Z will visit that state, and the system will fail to achieve PA with probability one. This can never happen in the deterministic setting, where the concentration of Z will always be nonzero for any nonzero initial condition.

In Briat et al. (2016a), the antithetic integral motif was proposed as a molecular integral feedback motif that exhibits particularly appealing features in the context of stochastic dynamics. It was shown that RPA will be achieved even when both controller species, Z_1 and Z_2, are fluctuating and have low copy numbers. In this case, RPA means that $\mathbb{E}(X_L) = \mu/\theta$ in spite of dynamic perturbations in the network and constant external inputs. Remarkably, it was also shown in Briat et al. (2016a) that RPA adaptation in the stochastic setting can be achieved even when the same system goes unstable in the noise-free deterministic case. Rather than being adversely affected by noise, the antithetic motif "exploits" it to achieve RPA, maintaining stability. In the case of stochastic dynamics, dynamic stability is taken to be closed-loop "ergodicity." This simply means that there exists a unique stationary distribution to which any initial distribution for the stochastic network will converge. The antithetic integral feedback motif achieves this feat by maintaining closed-loop ergodicity for a large class of stochastic networks interconnected with it, whereas in the deterministic setting, stability will not be maintained for the same class of networks.

As pointed out, RPA in the stochastic setting means $\mathbb{E}(X_L) = \mu/\theta$. This has two different interpretations. One is that the population average of $X_L(t)$ goes to μ/θ in the steady state. With this interpretation, nothing is said about the trajectory of individual cells, which will be fluctuating and will not converge in the usual sense. Another interpretation relates to single cells and says that in the steady state, the time average of $X_L(t)$ converges to μ/θ for each cell. More precisely,

$$\lim_{T\to\infty}\frac{1}{T}\int_0^T X_L(\tau)d\tau = \mu\Big/\theta.$$

For the antithetic integral feedback controller that achieves closed-loop ergodicity, both interpretations will be correct, and the population average as well as the single-cell time average will converge to the same value in the steady state.

We now consider a feedback interconnection of an arbitrary stochastic network, Σ, with a stochastic feedback network, Σ_F, as shown in Figure 8A. Given such an interconnection, we will say the interconnection achieves RPA if $\mathbb{E}(X_L) = \mu/\theta$ for all network perturbations of Σ and all constant disturbances. In what follows, we will assume the closed-loop system is ergodic. Under this condition, we can pose several fundamental questions: Is there a test to check that RPA is achieved by this interconnection? What architectural features are required for RPA? Can one parametrize all RPA-achieving feedback systems Σ_F, regardless of size? What is the minimal feedback system Σ_F that is guaranteed to achieve RPA? In Aoki et al. (2019), answers to all these questions were provided. We give a brief summary of these results here. RPA can be

characterized via a simple algebraic condition, which can be manipulated to reveal the internal structure of all RPA achieving Σ_F. In particular, every Σ_F that achieves RPA must embed within it an antithetic feedback motif. What this means is that the molecular species of every RPA-achieving Σ_F can be partitioned into three classes: $\mathcal{C}^+$, $\mathcal{C}^-$, and $\mathcal{C}^\emptyset$, and that there must exist one reaction that combines a species in the class $\mathcal{C}^+$ with a species in the class $\mathcal{C}^-$ to yield a product in the class $\mathcal{C}^\emptyset$, which also includes the empty set $\emptyset$ (see Figure 8B). This immediately implies that the minimal controller achieving RPA involves two species, Z_1 and Z_2, and is in fact the one given in Figure 5. In summary, these results give necessary and sufficient conditions for RPA in the stochastic setting and provide a parametrization of all feedback systems Σ_F that achieve RPA. While the antithetic integral feedback motif is a universal motif for RPA in the stochastic setting, it is not necessary for achieving RPA in the deterministic setting, but it is sufficient, as we have seen previously. See Aoki et al. (2019) for all the details.

It is interesting to note that the same stochastic antagonism that underlies the antithetic motif and is fundamental to its stochastic closed loop adaptation properties leads to large spontaneous fluctuations in open loop configurations. Such open-loop fluctuations are exploited to realize a different function: cell-fate-decision switching (Buchler and Cross, 2009; Elf et al., 2003; Lord et al., 2019).

RPA IN SYNTHETIC BIOLOGY

RPA has been the evolutionary strategy of choice in scenarios when tight robust regulation of biological variables is essential. Building synthetic circuits that realize RPA promises to deliver the same advantages that natural circuits enjoy, enabling variables of interest to be steered robustly toward tunable set points that are determined by the user. The ability of RPA to maintain these variables at their desired set points in spite of network uncertainty, dynamic perturbations, persistent disturbances, and stochastic noise makes it particularly suitable for synthetic biology, where the molecular players, their interactions, and the operating environment are rarely fully known.

In synthetic biology, much of the effort for RPA has focused on implementing integral feedback circuits. While synthetic negative feedback circuits were built and demonstrated as early as 2000, when the synthetic biology field itself was born, building integral feedback control circuits was particularly challenging. Some of these challenges were outlined in Ang and McMillen (2013), which explored the various physical constraints on biological integral control design for adaptation. Early attempts to mimic natural integral circuits were fraught with difficulties due to stringent tuning requirements needed to achieve zero-order reactions in the right dynamic range. However, the advent of the antithetic integral feedback control motif (Briat et al., 2016a) promised to avoid some of these steps. The key was to find molecules Z_1 and Z_2 that mutually annihilate, sequester, or inactivate one another in a stoichiometric fashion, while being stable in their free form. The need for stability of these molecules derives from the fact that a high degradation rate of these critical species destroys the integrator structure and introduces steady-state errors. In engineering, this is referred to as the "leaky" integrator effect. Large dilution rates due to cell growth could have the same effect (Ang and McMillen, 2013), although both effects could be mitigated by increasing the gain of the feedback loop. This ensures that the dynamics of the closed-loop network evolve at a fast timescale relative to the dilution rate of the controller species (Qian and Del Vecchio, 2018; Aoki et al., 2019).

These above-mentioned theoretical advancements set the stage for the synthetic realization of integral feedback circuits in living cells. Stable controller species that bind strongly with each other and can act as transcription factors may be realized by sigma/anti-sigma pairs that are commonly found in bacteria (Briat et al., 2016a). Using sigma factors from the bacterium *Bacillus subtilis*, the first synthetic integral feedback circuit demonstrated to achieve RPA in a living cell was first reported in 2017 (Lillacci et al., 2017; Aoki et al., 2019). The circuit, which implements the antithetic motif, is shown in Figure 9 and consists of 7 genes and 2 plasmids. Another circuit that achieved adaptation (in *E. coli*) used RNA instead of protein for the sequestration reaction (Huang et al., 2018a). As mRNA has a much shorter half-life than proteins in *E. coli*, stronger promoters needed to be used to overcome the effect of leaky integration, as mentioned above. Shortly thereafter, a circuit using sigma factors to implement the antithetic motif provided an *in vitro* demonstration of robust adaptation (Agrawal et al., 2019). More recently, integral feedback and RPA were demonstrated in mammalian cells (Frei et al., 2020). This circuit showed robustness to both external disturbances and dynamic perturbations. It relied on engineered sense/anti-sense mRNA for realizing the antithetic motif. Since mRNA is much more stable in mammalian cells than in bacteria, the effect of integral leakiness was not discernible over a range of operation. Incoherent feedforward circuits were also built synthetically to achieve adaptation. A recent example of such circuits (Segall-Shapiro et al., 2018) engineered IFFLs into *E. coli* promoters using transcription-activator-like effectors. Due to the adapting features of IFFLs, these promoters were shown to have near-identical expression in different genome locations and plasmids. Adaptation persisted despite changes in the promoter copy numbers and in composition of the growth medium.

Robust adaptation is only the first goal of implementing controllers in living cells. Indeed, adaptation is only a steady-state property and does not guarantee good transient performance. Recent theoretical work has focused on studying the dynamic response of the motifs that give RPA. In Briat et al. (2018) it was shown that adding a proportional feedback to the antithetic integral feedback motif enhances the dynamic performance of the circuit without impacting its adaptation properties. The resulting proportional-integral feedback circuit has better transient performance and reduced noise variance. In Chevalier et al. (2019) and Filo and Khammash (2021), designs based on derivative, proportional, and integral feedback were analyzed through computational and analytical methods and shown to improve transient performance. While such proportional-integral-derivative (PID) controllers are common in engineering practice, thus far they have not been realized in synthetic biology. An important consideration that faces circuit designers in engineering and biology alike is to understand the fundamental limitations and tradeoffs that exist in the design space. In Olsman et al. (2019a and 2019b), tools from control theory, particularly Bode's integral formula, were applied to study the antithetic integral motif and to characterize its stability and performance tradeoffs. This

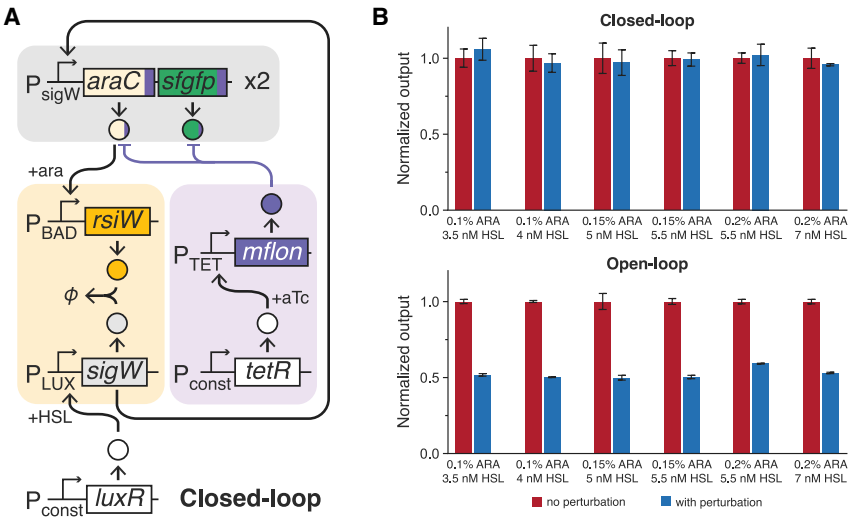


Figure 9. Antithetic integral feedback circuit in *E. coli*

(A) The earliest implementation of an antithetic integral control in a living cell (Lillacci et al., 2017; Aoki et al., 2019). A sigma and anti-sigma pair (rsiW and sigW) functioned as the controller species Z_1 and Z_2. The closed-loop circuit was used to robustly control the concentration of the protein AraC to a tunable setpoint determined by the concentration of the inducers HSL and arabinone, which set the parameters μ and θ, respectively. See Figure 5. The mflon system was used to test the controller's robustness to the protease mflon, which could be induced with aTc.

(B) The response of the circuit to constant disturbances (mflon) in both open- and closed-loop circuits. The antithetic circuit shows PA, completely rejecting the mflon disturbance in the steady state. In contrast, the open-loop circuit shows a large change in the output for the same disturbance.

work highlighted the intricate relationships between biomolecular rate parameters and system-level performance requirements, such as speed, adaptation error, and robustness, providing a good basis for understanding design tradeoffs in synthetic RPA networks employing the antithetic integral motif.

Controlling biological networks that are uncertain with precision and robustness despite noise, external disturbances, and/or dynamic perturbations opens up lots of opportunities for applications in industrial biotechnology and medical therapy. Early areas of application include biofuels (Briat and Khammash, 2018) and dynamic pathway regulation for cell factories (Boada et al., 2020), burden mitigation (Huang et al., 2018a), and growth control (Aoki et al., 2019). In medical therapy, RPA circuits may be used to restore homeostasis in the treatment of metabolic diseases, as well as for applications in immunotherapy and precise drug delivery.

CONCLUDING THOUGHTS

Mechanism plays a central and dominant role in biology. As mechanistic understanding has long been accepted as a basis for explaining biological function, the search for biological mechanism, with all its intricate molecular detail, has guided much of biological research. Given the sheer complexity of biological phenomena, elucidating all the molecular mechanisms supporting a particular function continues to be a daunting task, despite incredible advances in technology and methodology. Even when the search for parts and their interactions is painstakingly carried out over years of research, a clear overall picture of how the particulars of a mechanism contribute to the overall function may not emerge. Indeed, system-level patterns may not be visible at the molecular level, and the behavior of these patterns under different conditions even less so. At the same time, the same function may be realized using different mechanisms in different cell types, even when they achieve a similar goal. A different explanatory paradigm for biology may be possible. Rather than focusing exclusively on mechanism, it may be rewarding to look for structural constraints that all biological mechanisms that achieve a certain function must satisfy. Understanding these constraints and searching for motifs that respect them will inevitably lead to a deeper understanding of biological complexity. Even if one takes for granted that the goal is solely to uncover the biological mechanisms underlying a particular function, knowing *a priori* the constraints that such mechanisms must satisfy will guide and simplify the search for them. One can then understand what aspects of these mechanisms can be changed or evolved without impacting the function—something that would be extremely difficult to do just by knowing the molecular details. We have discussed in this perspective one biological function that lends itself to such an approach, namely RPA. We have seen that systems achieving RPA must be constructed in a specific way and that a small set of motifs is necessary and sufficient for RPA. The mechanistic details for realizing such RPA motifs may indeed be many and varied, relying on very different parts and interactions. But they all share the universality of the motifs that they comprise.

What other aspects of biological function lend themselves to this approach? And what other universal design principles can profitably be brought to bear to make sense of their complexity? It is quite conceivable, for example, that ideas from information theory and filtering theory can equally well constrain the design space for signaling pathways, adding a new dimension to the understanding of their design principles. Such approaches are indeed starting to be pursued, but existing theories, while powerful, will need to be further developed to better suit the biological context. This holds true for other theoretically rich fields interfacing with biology. With this in mind, it is not hard to imagine that a control theory, a filtering theory, and an information theory, suitably adapted for biological networks, can serve as the foundation for a new explanatory paradigm for biology. While this paradigm will not replace mechanistic understanding, it will give it color and meaning. Indeed, expanding and developing our understanding of structural biological constraints and the universal design principles from which they follow will afford us a new light with which we can see and understand biological organization—one that is every bit as illuminating as detail-oriented understanding.

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DECLARATION OF INTERESTS

The author is a member of the advisory board of *Cell Systems*. ETH Zurich has filed a patent application on behalf of the author and co-inventors of genetic circuit designs cited in this article.

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